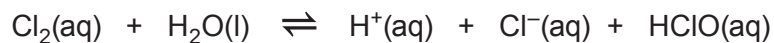


13. (a) During the 2016 Olympic games, the diving pool had to be closed after the water turned green. A report in the media incorrectly suggested that there was too much chlorine in the pool.

When chlorine gas dissolves in cold water, a pale green solution is formed. In this solution, the following equilibrium is established.



pale green

colourless

- (i) Chemical equilibria are often described as dynamic equilibria.

Explain the term *dynamic equilibrium*.

[1]

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- (ii) Use Le Chatelier's principle to explain why the pale green colour disappears if sodium hydroxide solution is added. [2]

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(b) Chlorine can react directly with metals to form chlorides.

- (i) I. Calcium chloride can exist as an anhydrous salt or as a hydrated salt,  $\text{CaCl}_2 \cdot x\text{H}_2\text{O}$ .

In an experiment to determine the extent of hydration a sample of hydrated calcium chloride,  $\text{CaCl}_2 \cdot x\text{H}_2\text{O}$ , with a mass of 3.29 g was heated to remove all water of crystallisation. The solid remaining had a mass of 1.67 g.

Calculate the value of  $x$  in the formula  $\text{CaCl}_2 \cdot x\text{H}_2\text{O}$ .

You **must** show your working.

[3]

$x = \dots\dots\dots$

- II. Suggest how a student doing this experiment would ensure that all the water had been removed. [1]

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.....

- (ii) The melting temperature of sodium chloride is 1074 K but sodium iodide has a melting temperature of 934 K.

Suggest why the melting temperature of sodium iodide is lower than that of sodium chloride. [1]

.....  
.....



(c) Chlorine forms molecules and ions with other halogens.

- (i) While chlorine has a boiling temperature of 238 K, the boiling temperature of iodine monochloride, ICl, is 371 K.

Suggest why the boiling temperature of iodine monochloride is higher than that of chlorine. [2]

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- (ii) Name the shape of the  $[\text{ClF}_6]^+$  ion. [1]

.....

- (iii)  $[\text{ClF}_2]^+$  and  $[\text{ClF}_2]^-$  are two other ions containing a chlorine atom.

A student said that their shapes must be different.

Is he correct? Justify your answer using VSEPR theory. [3]

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**END OF PAPER**



|               |               |                  |
|---------------|---------------|------------------|
| Surname       | Centre Number | Candidate Number |
| First name(s) |               | 2                |

**GCE AS/A LEVEL**

2410U10-1

**TUESDAY, 17 MAY 2022 – MORNING****CHEMISTRY – AS unit 1****The Language of Chemistry, Structure of Matter and Simple Reactions**

1 hour 30 minutes

**Section A****Section B****ADDITIONAL MATERIALS**

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

**INSTRUCTIONS TO CANDIDATES**

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

**Section A** Answer **all** questions.

**Section B** Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

The assessment of the quality of extended response (QER) will take place in **Q.10(a)**.

| For Examiner's use only |              |              |
|-------------------------|--------------|--------------|
| Question                | Maximum Mark | Mark Awarded |
| 1. to 7.                | 10           |              |
| 8.                      | 14           |              |
| 9.                      | 12           |              |
| 10.                     | 13           |              |
| 11.                     | 17           |              |
| 12.                     | 14           |              |
| <b>Total</b>            | <b>80</b>    |              |

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